

Event Management System — Functional Requirements (10)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow users to create an account using their name, email, and password. The user's email must be verified before the account is activated.	High	Pass: The account is created but remains inactive and cannot be used to log in until the user clicks the verification link sent by email. After verification, login succeeds. Fail: The account becomes active without email verification.	Ensures that user identities are verified before allowing login or role-based access.
FR-002	Functional	The system shall allow users to log in using their email and password and reset a forgotten password through an emailed reset link.	High	Pass: Valid credentials allow login. Selecting "Forgot Password" sends a time-limited reset link that successfully changes the password, making the old password invalid. Fail: Invalid credentials are accepted, or the reset link does not expire or fails to update the password.	Provides essential authentication and password recovery functionality.
FR-003	Functional	The system shall assign each user a role (Admin, Organizer, or Participant) and restrict access to features based on that role. Users shall be able to update their own profile information, while admins shall be able to view, suspend, or delete user accounts.	High	Pass: Each role can perform only its permitted actions; profile changes are saved correctly; suspended or deleted accounts are immediately prevented from logging in. Fail: A role can perform an unauthorized action, or a suspended account can still log in.	Maintains proper access control and allows administrators to manage user accounts.
FR-004	Functional	The organizer shall be able to create an event by providing its name, description, date, time, venue, category, and capacity. The organizer shall also be able to modify the event details after creation.	High	Pass: All required event details are stored correctly when an event is created, and subsequent changes are saved and displayed immediately. Fail: An event is created without a required field, or an update is not saved.	Provides the core functionality required for organizers to create and manage events.
FR-005	Functional	The organizer shall be able to delete or cancel an event. If an event that has registered participants is updated or cancelled, the system shall notify all affected participants.	High	Pass: A cancelled event no longer accepts registrations, and all registered participants receive a notification within 5 minutes. Fail: The event continues accepting registrations after cancellation, or any registered participant fails to receive the notification.	Ensures participants are promptly informed about changes that may affect their plans.
FR-006	Functional	Users shall be able to view available events and search by event name or filter events based on category, date, or venue. The admin shall be able to manage event categories.	High	Pass: The event list displays all upcoming, non-cancelled events. Search and filter results contain only relevant events, and category changes made by the admin are reflected throughout the system. Fail: Events are incorrectly included or excluded, or category changes are not updated properly.	Makes it easier for users to discover relevant events and ensures accurate search and filtering.
FR-007	Functional	A logged-in user shall be able to register for an event. The system shall store the user's registration details, including the user, event, timestamp, and status, while preventing duplicate registrations for the same event.	High	Pass: A valid registration creates exactly one correct record for the user-event combination, while a second registration attempt is rejected. Fail: Duplicate registrations are created or a valid registration is not recorded.	Ensures reliable event registration and maintains data integrity.

FR-008	Functional	The system shall verify the remaining event capacity before confirming a registration. Registrations shall be rejected when an event is full, and concurrent registration requests shall be handled without exceeding the available capacity.	High	Pass: Registration for a full event is rejected. For 100 simultaneous requests to a 1-seat event, exactly one request receives confirmation. Fail: A full event accepts a registration or multiple users are confirmed for the same final seat.	Prevents overbooking, which is one of the most critical failure scenarios for the platform.
FR-009	Functional	When an event reaches full capacity, users shall be given the option to join a waitlist. If a registered participant cancels, the system shall automatically offer the available seat to the next person on the waitlist. Users shall also be able to cancel their own registrations, and the system shall send registration, cancellation, and event-reminder notifications.	Medium	Pass: Waitlist entries are recorded in the correct order; when a seat becomes available, the first waitlisted user is automatically promoted and notified; self-cancellation updates the registration status and seat count immediately; required confirmation and reminder emails are sent within their specified time windows. Fail: An available seat remains unused despite a waitlisted user, or a required notification is not sent.	Maximizes registrations after cancellations and keeps users informed throughout the registration process.
FR-010	Functional	The system shall display the number of registered participants and available seats for each event. Organizers shall be able to view event-specific registration details and export participant lists in CSV or PDF format. Admins shall be able to view overall platform statistics, including total events, users, and registrations.	Medium	Pass: Displayed and exported counts accurately reflect current active registrations, and administrator statistics match the live database totals. Fail: Any displayed or exported information is outdated or inaccurate.	Provides organizers and administrators with the information needed to manage, monitor, and evaluate the platform.